Introduction

Classically mathematics would probably be divided into "Number Theory" and "Geometry". One side believed in the primacy of numbers as a way to abstract the world around us while the other side believed that shapes define our universe. Surprisingly, both of them are right! Numbers and Shapes interchangably provide the necessary abstraction that allows us to make sense of things.

Perhaps the earliest version of this is the fact that three people take a rope which is 12 units in length and hold it at distances 3, 4 and 5 units from each other in such a manner that the rope is taught, then one of the persons is sees the other two at a right angle. This is an expression of the Pythagoras theorem *as well as* the arithmetic result that $3^2 + 4^2 = 5^2$.

Another idea is to look at numbers that can be shaped into rectangles with non-trivial width and height. In other words, numbers n that can be written as $n = a \cdot b$ with $a, b > 1$. This is the same as the collection of *composite* numbers.

A slightly more advanced geometric question is the understand the "shape" of a sequence of numbers. For example, we can compare *triangular* numbers

$$1, 3, 6, 10, 15, 21, \ldots$$

with *Fibonacci* numbers

$$0, 1, 1, 2, 3, 5, 8, \ldots$$

It may not be immediately obvious, but the second sequence soon outgrows the first and eventually grows *much faster* than the first one.

Broadly, speaking we can thus divide the study of numbers into the following broad classes.

Algebraic Questions: Finding solutions (or more generally, the behaviour of solutions) of equations and inequalities formed using the standard arithmetic operations on numbers.

Special Numbers: Finding numbers that satisfy some specific properties or fitting some special pattern.

Function behaviours: Studying the behaviour of functions that give numbers.

In most cases, when we say "number" we mean counting numbers. However, there are natural extensions to negative numbers, rational numbers, and so on.

In this course, we will *primarily* focus on the *algebraic* aspect. However, the other aspects are not entirely separate and will play a role as well.

We now recall some topics from elementary number theory that will play an important role in our study.In what follows we will work with the collection $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ of natural or counting numbers.

Euclid's GCD Algorithm

Given two natural numbers $0 \leq a \leq b$, we carry out the following steps:

1. If $a = 0$ we declare output as b.
2. We use division to write $b = qa + r$ with $0 \leq r < a$.
3. We then replace a by r and b by a and go back to step (1).

Since $r < a$, one can see that we *must* output via step (1). We call this output as $\gcd(a, b)$.

Note that $r = qa - b$ at each stage.

Thus, any divisor of r and a divides b as well. It follows that the output divides a and b.

Moreover, if we *collect* the stages, we can write $\gcd(a, b) = ax - by$, with x and y allowed to be negative integers as well.

Note that *any* common divisor of a and b divides *each* of the numbers $ax + by$.

It follows that the output of this algorithm is the greatest common divisor of a and b.

Note that *every* number divides 0 so that $\gcd(0, b) = b$.

When $1 \leq a \leq b$, this means that $a = \gcd(a, b) \cdot a_1$ and $b = \gcd(a, b) \cdot b_1$ such that $\gcd(a_1, b_1) = 1$.

Details of proofs and much more can be found at the Wikipedia page on this algorithm.

Modular Arithmetic

We define the set $\mathbb{Z}/\langle N\rangle = \{0, 1, \ldots, N - 1\}$.

On this set we have the operations of addition and multiplication modulo N with 0 and 1 serving as additive and multiplicative identity.

This makes $\mathbb{Z}/\langle N\rangle$ a ring with (multiplicative) identity in which multiplication is also commutative.

One of the objects of interest is the group $U(\mathbb{Z}/\langle N\rangle)$ of *invertible* elements of this ring. That is to say, numbers a such that $0 < a < N$ such that there is a number b for which $a \cdot b - 1$ is divisible by N. Note that this means $\gcd(a, N) = 1$. Conversely, if we use Euclid's GCD algorithm to calculate $\gcd(a, N)$ and we get output 1, then we have written $ax + Ny = 1$, so we can take $b = (x \mod N)$.

$$U(\mathbb{Z}/\langle N\rangle) = \{a : 0 < a < N \text{ and } \gcd(a, N) = 1\}$$

The Euler *totient* function $\varphi(N)$ is defined as the *size* of this set.

Details of proofs and much more can be found at the Wikipedia page on Modular Arithmetic.

Chinese Remainder Theorem

If M divides N, then we have a map $\mathbb{Z}/\langle N\rangle \to \mathbb{Z}/\langle M\rangle$ by sending a to $(a \mod M)$. This map preserves addition, multiplication, 0 and 1; in other words, it is a homomorphism of rings.

It follows that we have a map

$$\mathbb{Z}/\langle M \cdot N\rangle \to \mathbb{Z}/\langle M\rangle \times \mathbb{Z}/\langle N\rangle$$

The question we look at is whether this map is *onto*. Since both sides have the same number of elements, this is the same as asking whether this map is *one-to-one*.

Given r such that $0 \leq r < M$, and s such that $0 \leq s < N$, we look at the problem of finding m and n such that $r + mM = s + nN$. This means that $r - s = nN - mM$ is divisible by $\gcd(M, N)$.

In particular, if $\gcd(M, N) \neq 1$, we *can* find r and s such that $r - s$ is *not* divisible by it (for example $r = 1$ and $s = 0$). Thus the map is *not* onto.

If $\gcd(M, N) = 1$, then we can write $Mp - Nq = 1$ for some p and q integers.

It follows that $Mp(r-s) - Nq(r-s) = r - s$. It follows that $r - Mp(r-s) = s - Nq(r-s) = t$. We then note that $(t \mod MN)$ is an element lf the left-hand side that has image (r, s) in the right-hand side.

Details of proofs and much more can be found at the Wikipedia page on Chinese Remainder Theorem.

Primality and Factorisation

A number $p > 1$ is prime if one of the following *equivalent* conditions holds:

- For every a, $\gcd(p, a)$ is either 1 or p.
- Given $p = a \cdot b$ either a or b is 1.
- Given p divides $a \cdot b$, then either p divides a or p divides b.

Similarly, we can say that a number $q > 1$ is a prime power if one of the following *equivalent* conditions holds:

- For every a, either $\gcd(q, a)$ is 1 or q and $\gcd(q, a)$ have a common power.
- Given $q = a \cdot b$ with $a, b > 1$ we have $\gcd(a, b) \neq 1$.
- Given q divides $a \cdot b$, then either q divides a or q divides a power b^n of b.

Every number N can be *uniquely* written as a product of prime powers in the form

$$N = p_1^{a_1} \cdots p_k^{a_k}$$

where $p_1 < \cdots < p_k$ and $a_i > 0$.

Details of proofs and much more can be found at the Wikipedia page on Prime Numbers and the Wikipedia page on Prime Powers.

Sums of multiples

An interesting exercise (which is useful to try to work out on one's own!) is as follows.

Given a and b which have $\gcd(a, b) = 1$ and look at the set of all numbers of the form $ap + bq$ where $p, q \geq 0$.

- Show that *every* sufficiently large number is of this form.
- Find a formula for the largest number which is *not* of this form.